HAIRUL AZHAR BIN ABU BAKAR

FORWARD-DEPLOYED AI ENGINEER · AI SOLUTIONS ARCHITECT · FOUNDER

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GitHub github.com/CybergigSG

Forward-deployed AI engineer and founder who builds agentic AI and runs it in production. I architect **KatalockAi**, a live multilingual AI workspace localised for India, Singapore and Indonesia; its **523-lesson Academy** with an AI tutor that reviews student code; and **AIM OS**, a governed platform for autonomous agents. KatalockAi has shipped live client products, including an AI tarot-reading app and an investor website. Before AI, I spent 10+ years accountable for budgets, SLAs and on-site delivery across integrated-resort fit-outs, luxury-hospitality supply and enterprise logistics. That is why I build AI the way operators need it: verified, gated and recoverable.

LIVE PRODUCTS & CLIENT DELIVERY

KatalockAi
www.katalock.net

Multilingual agentic workspace: tiered model stack, K-Computer cross-model verification, gated browser automation, live code sandbox.

KatalockAi Academy
www.katalock.net/academy

20 phases, 523 lessons, 6 career routes. Professor Katalock tutors and grades projects against rubrics.

AIM OS
aimglobal.io

Governed agent OS: 50+ apps, 108 API endpoints, 27+ agents, QGov policy engine and killswitch.

Tarot with Shilpa
www.tarotwithshilpa.com

Client AI product: reader personas, multilingual readings, freemium metering, sign-in, hand-off to a human reader.

Ignis Eco
www.igniseco.com

Investor-facing site for a plastic-to-fuel venture: process explainer, plant overview, language switcher.

AI video · virtual staging
Showreel on portfolio

18-second 1080p generation: an empty room furnished frame by frame, for property marketing.

TECHNICAL SKILLS

AI & agents	Agentic AI · Multi-agent orchestration · Cross-model verification · RAG · Model Context Protocol (MCP) · Knowledge distillation · Quantization · Evaluation & release gates · Anti-hallucination governance · Pinecone · Chroma
LLM serving (taught)	vLLM · SGLang · TensorRT-LLM · KV-cache optimisation · AWQ / GPTQ / FP8 · Speculative decoding · RLHF / DPO
Full stack & infra	Python · TypeScript · React / Next.js · Node.js / Express · FastAPI / Django · PostgreSQL / Supabase · Redis · Docker / Kubernetes · AWS / GCP · Terraform · Stripe
Delivery	Workflow discovery · Stakeholder management · SLA operations · Budget ownership ($500K–$5M) · Go-to-market · Investor relations

PROFESSIONAL EXPERIENCE

AI Platform Architect & Co-Founder | KatalockAi
2026 – Present

Global platform · Remote

- Own backend architecture, agent orchestration and infrastructure end to end for a global agentic AI workspace; co-founder runs on-the-ground operations.
- Built a tiered model stack (katalock-lucy, -tara, -stargazer, -aim) using **knowledge distillation and cost-performance routing**, localised in model, language and currency for India, Singapore and Indonesia.
- Built **K-Computer**, which runs independent model passes, reconciles disagreements and returns one answer with a confidence level.

- Shipped supervised browser automation that pauses for approval before any submit or payment, plus live workflow dependency graphs, an evidence-ledger CRM, and a code sandbox that builds, runs, publishes and pushes to GitHub.
- Hardened the platform for real use: streamed heartbeats to remove timeouts, and an atomic, server-authoritative credit wallet secured by row-level security.

Client Delivery | KatalockAi **2026 – Present**
Part of the KatalockAi role · Singapore / India

- Delivered **Tarot with Shilpa**, a live AI reading product: persona-based readers, spread selection, multilingual answers, freemium metering of 3 free questions a month, sign-in, reviews and hand-off to a human reader.
- Built and launched the **Ignis Eco** investor website: problem narrative, pyrolysis process explainer, plant overview, team and FAQ, with a language switcher.
- Packaged done-for-you AI for small and mid-sized businesses (24/7 multilingual voice desks, social storefronts, automated hiring) into bundles for clinics, retail, real estate and restaurants.
- Produced AI video for property marketing, including virtual staging that turns an empty room into a furnished interior in a single 1080p generation.

Architect & Founder | KatalockAi Academy **2026 – Present**
Global platform · Remote

- Built the learning platform end to end: **20 phases, 523 lessons, 6 career routes, 6 skill paths and 17 certified capstones**, with phase exams and certificates carrying verifiable IDs.
- Built **Professor Katalock**, a Socratic AI tutor embedded in every lesson. It checks understanding, reviews student code, flags edge cases, grades projects against rubrics, and reveals solutions only after an attempt.
- Designed the curriculum from first-principles maths to production agents. It covers LLM serving (vLLM, SGLang, TensorRT-LLM, PagedAttention, prefill/decode disaggregation), quantization economics (AWQ, GPTQ, FP8, NVFP4, speculative decoding) and alignment (SFT, RLHF, DPO/IPO/KTO, Constitutional AI).
- Mapped career routes to current job descriptions, including Forward-Deployed AI Engineer, Agent Systems Engineer and AI Evaluation Engineer; added MCP and Claude certification prep tracks.

Founder & AI Platform Architect | AIM OS (aimglobal.io) **Jan 2023 – Present**
Singapore · Remote

- Architected AIM OS, an operating system for agentic AI spanning 50+ integrated apps, 108 API endpoints, 27+ autonomous agents and about 80 database tables.
- Designed the governance layer: **QGov** policy engine, **Sentinel** safety monitor and a founder killswitch that keep agents inside defined boundaries.
- Built **Nexus Hub** orchestration with intent routing, a pg-boss async job queue with automatic retries, and offline-first real-time collaboration on Yjs CRDTs.

Co-Founder, Investor Relations & GTM | Ignis Eco (igniseco.com) **2026 – Present**
Tamil Nadu / Bengaluru, India

- Lead investor relations, seed-stage capital structuring, go-to-market and stakeholder outreach for a plastic-to-fuel pyrolysis venture.

Logistics Technology Operations (contract) | GOGOVAN Singapore (GOGOX) **Jul 2024 – Sep 2024**
Singapore

- Ran enterprise logistics technology for IKEA Singapore: **99.8% on-time delivery** across 5,000+ daily orders, with incident resolution time cut by 25%.

Co-Founder & Operations Director | Freda'D Pte Ltd / DGCC Investment Pte Ltd **Apr 2014 – Dec 2024**
Singapore / Middle East · run concurrently with employment roles

- Grew a bootstrapped fragrance business from $500 to a profitable international company supplying luxury hotel portfolios in Asia and the Middle East, at **25% YoY growth**.
- Deployed automation and inventory systems in client operations, cutting overhead by 30% and lifting customer engagement by 40% while holding 99.8% on-time delivery.

Project Director | HH Design Pte Ltd / Interior Producer Pte Ltd **Jan 2010 – Mar 2020**
Singapore

- Delivered integrated-resort fit-outs including **Marina Bay Sands Casino** and **Resorts World Sentosa**, owning $500K–$5M budgets and improving timelines by 15%.

FEATURED PROJECTS

1. KatalockAi · Multilingual Agentic Workspace www.katalock.net
Next.js, React, Supabase/PostgreSQL, distillation pipelines, Composio, Docker, Stripe
Live workspace with tiered model stack, K-Computer verification, gated browser automation and a code sandbox, localised for three markets.

2. KatalockAi Academy & Professor Katalock www.katalock.net/academy
Next.js, React, Supabase, automated evaluation, rubric grading
523-lesson AI engineering platform with an AI tutor that reviews code and grades projects; certificates with verifiable IDs.

3. Tarot with Shilpa · Client AI Product www.tarotwithshilpa.com
Built with KatalockAi: LLM personas, multilingual UI, auth, usage metering
Consumer AI reading app with selectable reader personas and spreads, a monthly free tier, reviews and escalation to a human reader.

4. AIM OS · Governed Agent Operating System aimglobal.io
Python, React, Node.js, PostgreSQL, Redis, Docker, Kubernetes, vector DBs
Agent OS with QGov governance, Sentinel safety monitoring and Nexus Hub multi-agent orchestration.

5. WorkflowGenius · Enterprise Document Automation github.com/CybergigSG/WorkflowGenius
Python, RAG, Pinecone, React, AWS Lambda/EC2
RAG pipeline processing 10,000+ daily transactions at 99.5% uptime on AWS, reducing administrative workload by 40%.

EDUCATION

MSc Computer Science
Heriot-Watt University · expected May 2028

Diploma in Architectural Technology
Singapore Polytechnic · 2001–2004

CERTIFICATIONS

IBM Full Stack Software Developer
Professional Certificate, 2024 · React, Node.js, Python, Django, Docker, Kubernetes, OpenShift

Google AI & Machine Learning
Professional Certificate, 2024 · GenAI for software development, Python data science

Google Cybersecurity
Professional Certificate, 2024

ADDITIONAL INFORMATION

Languages: English (native), Malay (native)
Work authorisation: Singapore citizen. Open to relocation: USA, UK, EU, UAE, Australia, Canada, India, APAC.
Availability: Immediate, or two weeks' notice.